

# Companion to Bird & de Moor, Algebra of Programming

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \multimap A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \multimap PA$  (8.1).

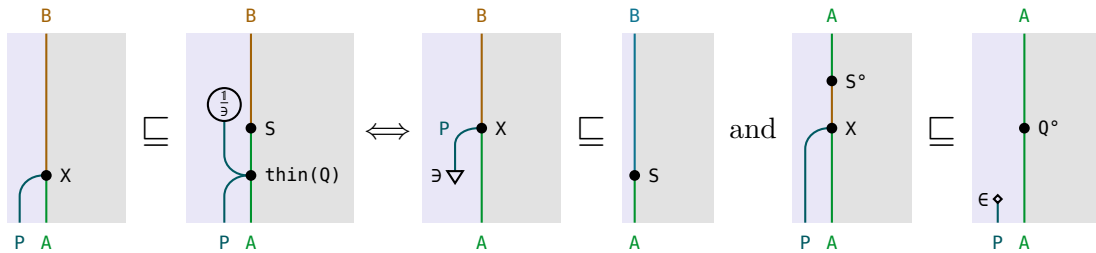
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \multimap ys \multimap xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \multimap b \multimap xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

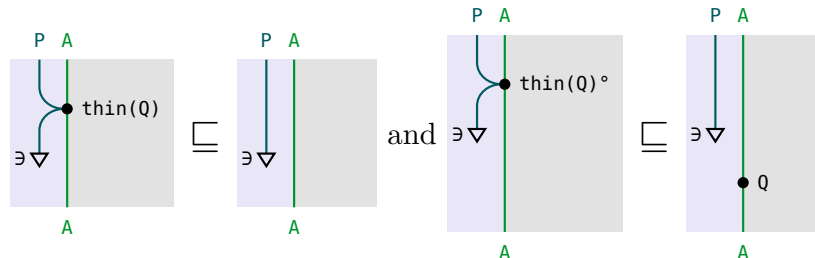
| the law                                                                                                                                                                                                                   | what it says                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$                                                                                                 | $S$ is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |
| 2 $\text{thin}(Q) \ni \sqsubseteq \exists$                                                                                                                                                                                | everything a thinning keeps was in the set                                                                                                                                           |
| 3 $\text{thin}(Q)^\circ \ni \sqsubseteq \exists \ Q$                                                                                                                                                                      | every element of the set has a $Q$ -lower bound among the kept ones                                                                                                                  |
| 4 $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$                                                                                                                                                    | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                                                                                    |
| 5 $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$<br>$Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$                                                                    | keeping everything is always a legal thinning                                                                                                                                        |
| 6 $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$<br>$1 \sqsubseteq Q$<br>$\implies \text{est}(R) \sqsubseteq \text{thin}(Q) \ \text{est}(R)$ | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                                                                                |
| 7 $\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists} \quad (8.2) - (8.2) \ \text{est}(Q) \ \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$                                                                       | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on                                                                |
| 8 $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant                                              | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                                                                                   |
| 9 $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4) - (8.4)$<br>$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$                                                            | thinning each member set is a thinning of the union                                                                                                                                  |

#### §8.1.1. $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)

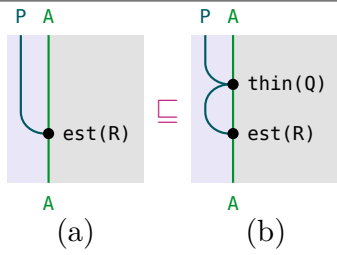


$X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$  at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ , since  $\frac{\exists}{\exists} = \wedge(\exists) = 1$

**§8.1.2.**  $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$  — thinning first, then taking the smallest, is the same as taking the smallest

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

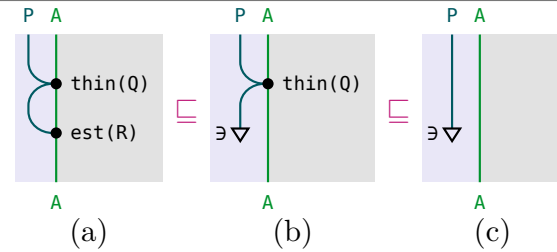
the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$



|     |                                             |                                                                                                                                                                                                               |
|-----|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{est}(R)$                             | $1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$<br>$\Rightarrow 1 \sqsubseteq \text{thin}(Q)$ ,<br>$Q \sqsubseteq Q$<br>$\Rightarrow \text{thin}(Q)$<br>$\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ |
| (b) | $\sqsubseteq \text{thin}(Q) \text{ est}(R)$ |                                                                                                                                                                                                               |

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

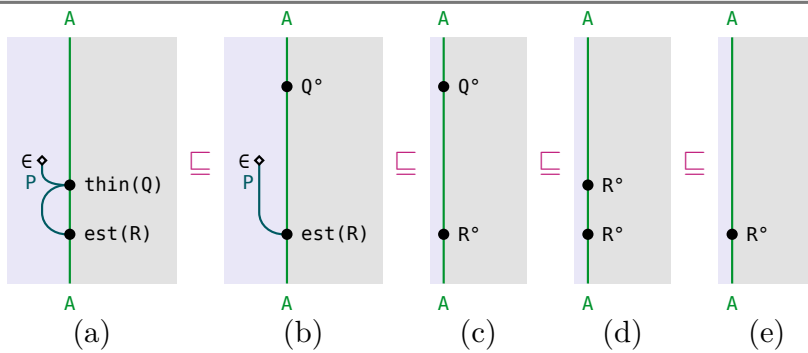
the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



|     |                                      |                                              |
|-----|--------------------------------------|----------------------------------------------|
| (a) | $\text{thin}(Q) \text{ est}(R)$      | $R \cap S \sqsubseteq R$                     |
| (b) | $\sqsubseteq \text{thin}(Q) \exists$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \exists$                |                                              |

$$Q \sqsubseteq R \wedge R R \sqsubseteq R \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the  $\sqsubseteq$  half, second condition



|     |                                         |                                                                          |
|-----|-----------------------------------------|--------------------------------------------------------------------------|
| (a) | $\in \text{thin}(Q) \text{ est}(R)$     | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$                             |
| (b) | $\sqsubseteq Q^\circ \in \text{est}(R)$ | $\in \text{est}(R) \sqsubseteq R^\circ$                                  |
| (c) | $\sqsubseteq Q^\circ R^\circ$           | $Q \sqsubseteq R$                                                        |
| (d) | $\sqsubseteq R^\circ R^\circ$           | $R R \sqsubseteq R$<br>$\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$ |
| (e) | $\sqsubseteq R^\circ$                   |                                                                          |

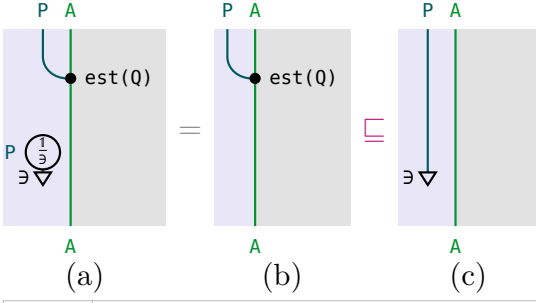
$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

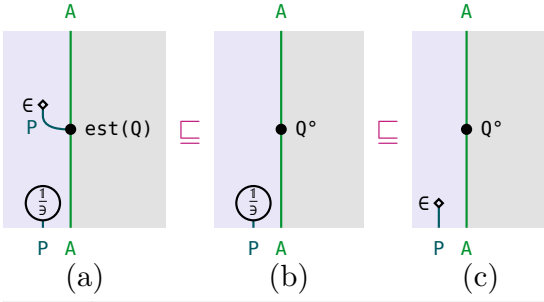
every member of the singleton is a member of the set —  $\triangleq \mathbf{thin}, \ni/\ni$  half



|     |                                             |                                 |
|-----|---------------------------------------------|---------------------------------|
| (a) | $\mathbf{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ |
| (b) | $= \mathbf{est}(Q)$                         | $R \cap S \sqsubseteq R$        |
| (c) | $\sqsubseteq \ni$                           |                                 |

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every member of the set —  $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$  half

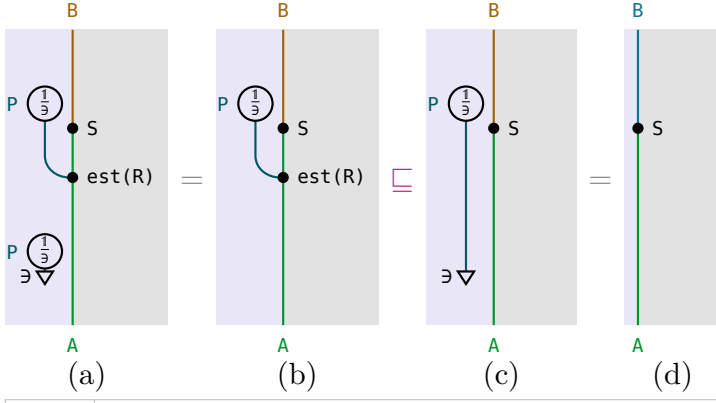


|     |                                             |                                           |
|-----|---------------------------------------------|-------------------------------------------|
| (a) | $\in \mathbf{est}(Q) \stackrel{1}{\exists}$ | $\in \mathbf{est}(R) \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq Q^\circ \stackrel{1}{\exists}$ | $\stackrel{1}{\exists} \sqsubseteq \in$   |
| (c) | $\sqsubseteq Q^\circ \in$                   |                                           |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

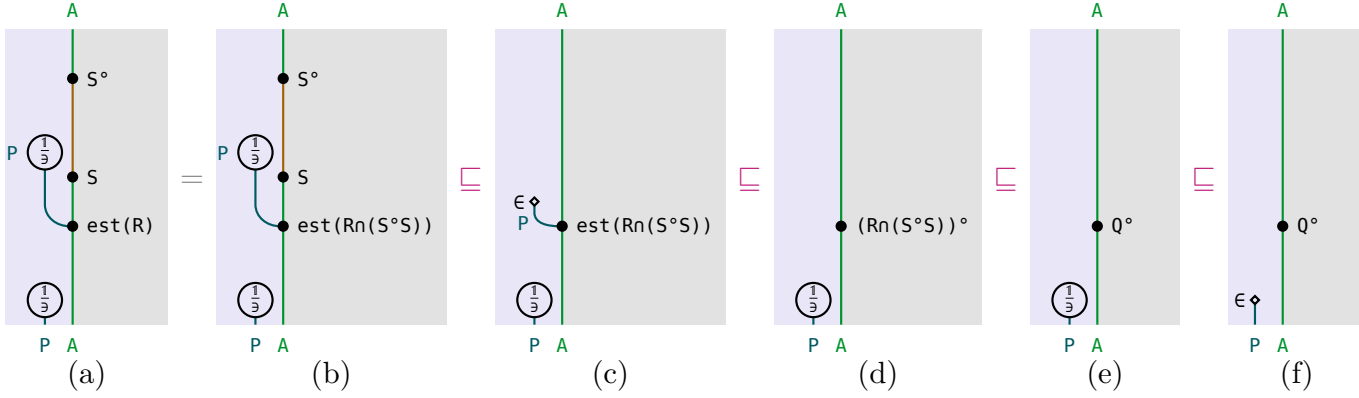
every member of the singleton is a value  $S$  returns —  $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , first condition



|     |                                                         |                             |
|-----|---------------------------------------------------------|-----------------------------|
| (a) | $\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$ | $\frac{1}{\exists} \ni = 1$ |
| (b) | $= \frac{S}{\exists} \text{est}(R)$                     | $R \cap S \subseteq R$      |
| (c) | $\subseteq \frac{S}{\exists} \ni$                       | $\frac{R}{\exists} \ni = R$ |
| (d) | $= S$                                                   |                             |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument —  $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , second condition



|     |                                                                                |                                                                                      |
|-----|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$                    | $\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$ |
| (b) | $= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$ | $T^\circ \frac{1}{\exists} \subseteq \in$                                            |
| (c) | $\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$               | $\in \text{est}(R) \subseteq R^\circ$                                                |
| (d) | $\subseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$                       | $R \cap (S^\circ S) \subseteq Q$                                                     |
| (e) | $\subseteq Q^\circ \frac{1}{\exists}$                                          | $\frac{1}{\exists} \subseteq \in$                                                    |
| (f) | $\subseteq Q^\circ \in$                                                        |                                                                                      |

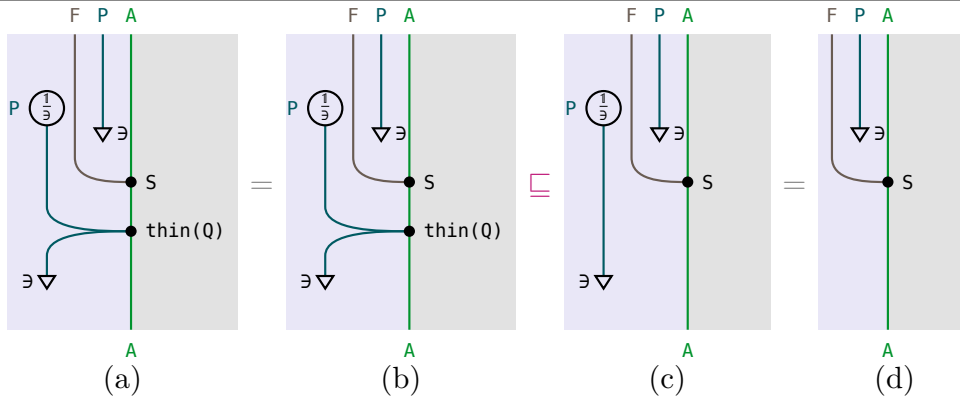
$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

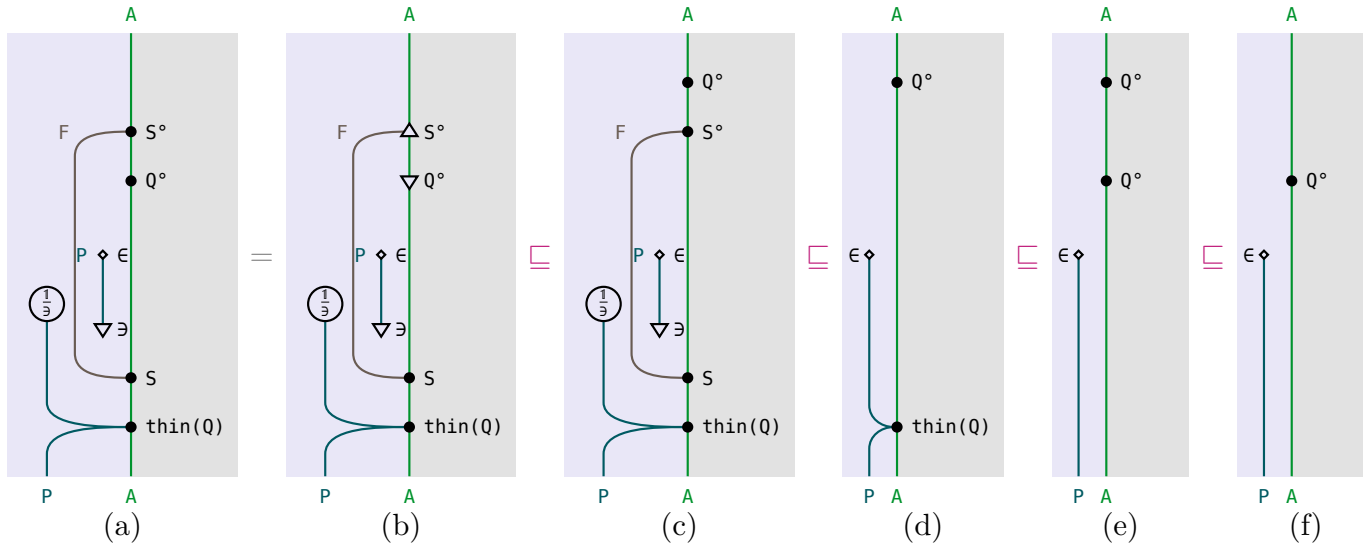
everything the thinning fold keeps is a value of the plain fold —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , first condition, by fusion from the chain below



|     |                                                     |                                          |
|-----|-----------------------------------------------------|------------------------------------------|
| (a) | $\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$   |                                          |
| (b) | $= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$ | $\text{thin}(Q) \ni \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \frac{F(\exists) S}{\exists} \ni$      | $\frac{R}{\exists} \ni = R$              |
| (d) | $= F(\exists) S$                                    |                                          |

$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is  $Q$ -above one the thinning fold keeps —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , second condition, by hylomorphism theorem from the chain below



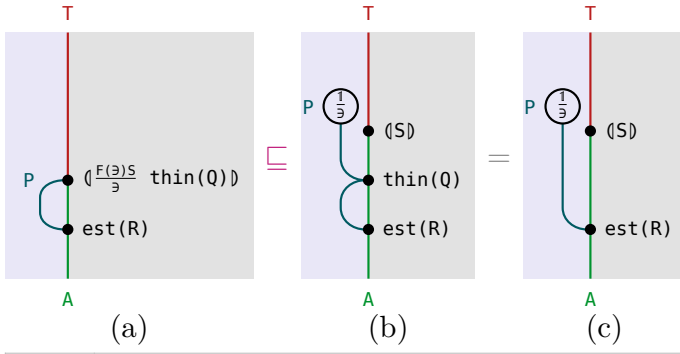
|     |                                                                                       |                                                                         |
|-----|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| (a) | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$                  |                                                                         |
| (b) | $= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$        | $F(Q) S \sqsubseteq S Q$                                                |
| (c) | $\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$               |
| (d) | $\sqsubseteq Q^\circ \in \text{thin}(Q)$                                              | $S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$ |
| (e) | $\sqsubseteq Q^\circ Q^\circ \in$                                                     | $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$                       |
| (f) | $\sqsubseteq Q^\circ \in$                                                             |                                                                         |

(8.1.2d)

| step | type                                                                                                                                             | example                | what it does                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------|
| 1    | $S: F(A) \rightarrow A$ is 8.1's algebra; example $F(X) = V + V \times X$ , $S = [\text{wrap}, \text{cons}]$ , candidate $a = \text{cons}(v, p)$ |                        |                                                                                          |
| 2    | $S^\circ$                                                                                                                                        | $A \rightarrow FA$     | $\text{cons}(v, p) \mapsto r(v, p)$                                                      |
| 3    | $F(Q^\circ \in)$                                                                                                                                 | $FA \rightarrow F(PA)$ | $r(v, p) \mapsto r(v, ps)$ , some $p' \in ps$ with $p' \sqsubseteq p$                    |
| 4    | $\frac{F(\exists) S}{\exists}$                                                                                                                   | $F(PA) \rightarrow PA$ | $r(v, ps) \mapsto \{\text{cons}(v, t) \mid t \in ps\}$                                   |
| 5    | $\text{thin}(Q)$                                                                                                                                 | $PA \rightarrow PA$    | $\{\text{cons}(v, t) \mid t \in ps\} \mapsto ys$                                         |
| 6    | $\sqsubseteq Q^\circ \in$                                                                                                                        | $A \rightarrow PA$     | $ys$ holds $y$ with $y \sqsubseteq \text{cons}(v, p') \sqsubseteq \text{cons}(v, p) = a$ |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) S \sqsubseteq S Q \implies (\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{\exists} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



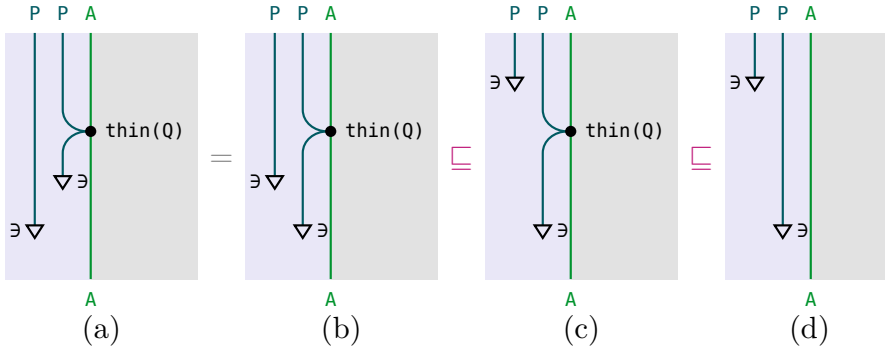
|     |                                                                          |                                                                                                                                                                                      |
|-----|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{est}(R)$ | Theorem 8.1 $\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q$<br>$\implies (\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q)$ |
| (b) | $\sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \text{est}(R)$           |                                                                                                                                                                                      |
| (c) | $= \frac{(S)}{\exists} \text{est}(R)$                                    |                                                                                                                                                                                      |

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists \exists$$

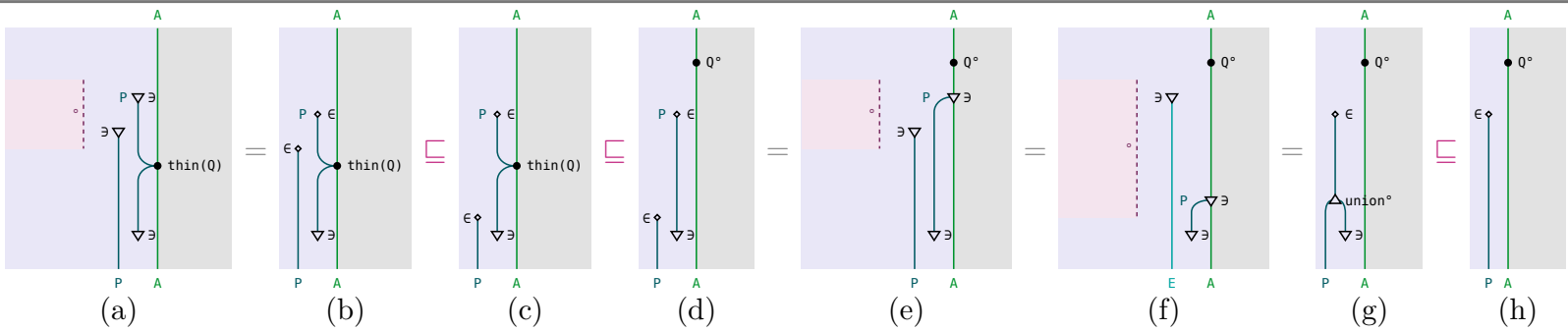
every member of the result is a member of a member set —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , first condition



|     |                                              |                                              |
|-----|----------------------------------------------|----------------------------------------------|
| (a) | $P(\text{thin}(Q)) \text{ union} \exists$    | $\text{union} \exists = \exists \exists$     |
| (b) | $= P(\text{thin}(Q)) \exists \exists$        | $P(R) \exists \sqsubseteq \exists R$         |
| (c) | $\sqsubseteq \exists \text{thin}(Q) \exists$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (d) | $\sqsubseteq \exists \exists$                |                                              |

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a  $Q$ -lower bound in the result —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , second condition



|     |                                                           |                                              |
|-----|-----------------------------------------------------------|----------------------------------------------|
| (a) | $(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$ | $(R S)^\circ = S^\circ R^\circ$              |
| (b) | $= \in \in P(\text{thin}(Q)) \text{ union}$               | $\in P(R) \sqsubseteq R \in$                 |
| (c) | $\sqsubseteq \in \text{thin}(Q) \in \text{union}$         | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ |
| (d) | $\sqsubseteq Q^\circ \in \in \text{union}$                | $(R S)^\circ = S^\circ R^\circ$              |
| (e) | $= Q^\circ (\exists \exists)^\circ \text{union}$          | $\text{union} \exists = \exists \exists$     |
| (f) | $= Q^\circ (\text{union} \exists)^\circ \text{union}$     | $(R S)^\circ = S^\circ R^\circ$              |
| (g) | $= Q^\circ \in \text{union}^\circ \text{union}$           | $\text{union} = E(\exists)$                  |
| (h) | $\sqsubseteq Q^\circ \in$                                 |                                              |

## §8.2. Paths in a layered network

(8.2a)

|    | name                            | type                                                               | definition                                                                             | meaning                                                                                                                                                          |
|----|---------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | F                               | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$           | $F(B, X) = B + (B \times X)$                                                           | a path is either one vertex $a$ , or a vertex $a$ followed by a path $xs$                                                                                        |
| 2  | $\alpha$                        | $F(A, LA) \rightarrow LA$                                          | $\alpha = [\text{wrap}, \text{cons}]$                                                  | build a path: wrap one vertex as $[a]$ , or put $a$ in front of $xs$                                                                                             |
| 3  | wrapz                           | $A \rightarrow LA \times \mathbb{N}$                               | $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$ | a path of one vertex, no edge, cost 0                                                                                                                            |
| 4  | consw                           | $A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$ | $(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$                  | put $a$ in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$                                                           |
| 5  | cost                            | $LA \rightarrow \mathbb{N}$                                        | $\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \circ \pi_2$                    | the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$                                                                       |
| 6  | R                               | $LA \rightarrow LA$                                                | $R \triangleq \text{cost} \leq \text{cost}^\circ$                                      | $xs$ is cheaper than $ys$                                                                                                                                        |
| 7  | $L(R)$                          | $LA \rightarrow LB$                                                | $([F(\exists, 1) \alpha]) = L(\exists)$                                                | $L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer                                                                      |
| 8  | minpath                         | $L(PA) \rightarrow LA$                                             | $\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$                  | the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one |
| 9  | Q                               | $LA \rightarrow LA$                                                | $Q \triangleq R \cap (\text{head} \text{ head}^\circ)$                                 | like $R$ , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$                                                                  |
| 10 | l                               | $B \rightarrow B + C$                                              | $a \mapsto l(a)$                                                                       | mark a value as the left alternative                                                                                                                             |
| 11 | r                               | $C \rightarrow B + C$                                              | $b \mapsto r(b)$                                                                       | mark a value as the right alternative                                                                                                                            |
| 12 | cpl                             | $PB \times A \rightarrow P(B \times A)$                            | $\text{cpl} \triangleq \text{cp}(1 \times A)$                                          | pair every element of the set with the value beside it                                                                                                           |
| 13 | cpr                             | $A \times PB \rightarrow P(A \times B)$                            | $\text{cpr} \triangleq \text{cp}(A \times 1)$                                          | pair the value beside the set with every element of the set                                                                                                      |
| 14 | $\frac{F(\exists, 1)}{\exists}$ | $F(PB, X) \rightarrow PF(B, X)$                                    | $\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$                             | cpl transpose: from layer $x$ and a set of paths $ps$ , choose a vertex $a \in x$                                                                                |
| 15 | $\frac{F(1, \exists)}{\exists}$ | $F(B, PX) \rightarrow PF(B, X)$                                    | $\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$           | cpr transpose: from a vertex $a$ and a set of paths $ps$ , choose a tail $xs \in ps$                                                                             |
| 16 | step                            | $A \times P(LA) \rightarrow LA$                                    | $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$                      | put $a$ in front of every path in $ps$ and keep a cheapest one                                                                                                   |

(8.2b)

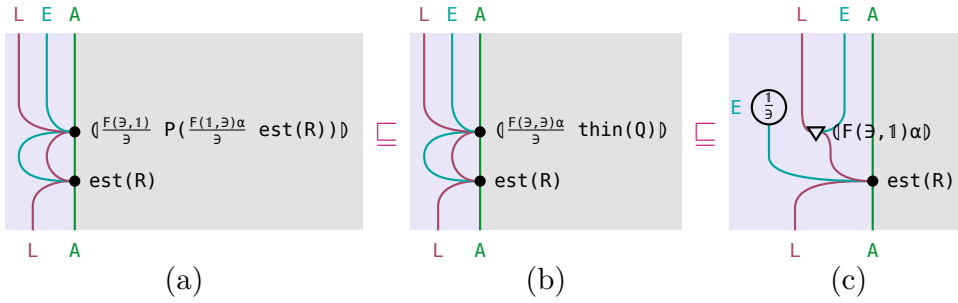
|   | the law                                                                         | what it says                                                                                                                                                             |
|---|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                             |
| 2 | $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

(8.2a)

$$\left( \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \right) \text{ est}(R) \sqsubseteq \left( \frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R) \quad (8.2c)$$

a least-cost path in a layered network, as a fold over the layers

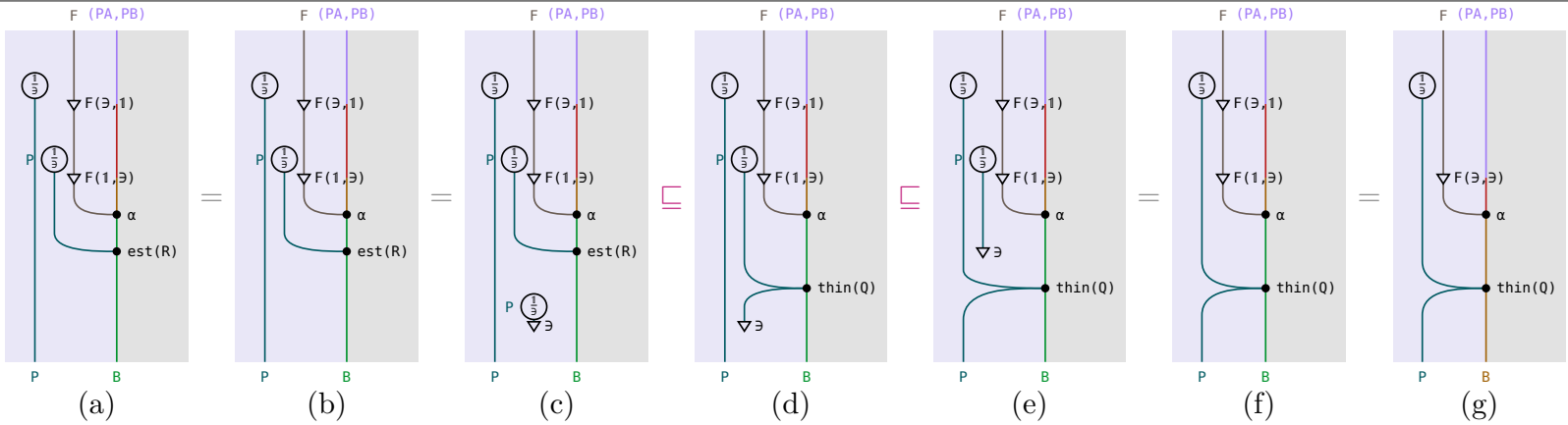
- Why it is needed:  $\text{est}(R)$  keeps one cheapest  $w$  of the set  $\Lambda(S)(x)$ ;  $\text{thin}(Q)$  needs every dropped  $z$  to be  $Q$ -beaten by a kept one. For  $\{w\}$  alone to be a thinning we need  $w \leq Q \leq z$  for every  $z$  in  $\Lambda(S)(x)$ .
- We have  $w \leq R \leq z$  ( $w$  is cheapest) and  $w \leq (S^\circ S) \leq z$  (both are built by  $S$  from the same argument  $x$ ); the hypothesis turns the two into  $w \leq Q \leq z$ : among candidates from one argument, cheaper already means  $Q$ -better.



|     |                                                                                                                               |                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\left( \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \right) \text{ est}(R)$ | $\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \sqsubseteq \frac{F(\exists,\exists)}{\exists} \text{ thin}(Q)$          |
| (b) | $\sqsubseteq \left( \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \right) \text{ est}(R)$                         | $\left( \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \right) \text{ est}(R) \sqsubseteq \left( \frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R)$ |
| (c) | $\sqsubseteq \left( \frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R)$                                               |                                                                                                                                                                         |

$$S \triangleq F(1,\exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \sqsubseteq \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



|     |                                                                                                                                |                                                                                                                                                              |
|-----|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right)$                                |                                                                                                                                                              |
| (b) | $= \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right)$                              | $P\left(\frac{1}{\exists}\right) E(\exists) = 1$                                                                                                             |
| (c) | $= \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R) \frac{1}{\exists}\right) E(\exists)$ | (8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$<br>$\Rightarrow \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{ thin}(Q)$ |
| (d) | $\sqsubseteq \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \right) P(\text{thin}(Q)) E(\exists)$     | (8.4) $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$                                                                                     |
| (e) | $\sqsubseteq \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \right) E(\exists) \text{ thin}(Q)$       | $\frac{V S}{\exists} = \frac{V}{\exists} P\left(\frac{S}{\exists}\right) \text{ union}$                                                                      |
| (f) | $= \frac{F(\exists,1)}{\exists} \frac{F(1,\exists)}{\exists} \alpha \text{ thin}(Q)$                                           | $F(R,1) F(1,S) = F(R,S)$                                                                                                                                     |
| (g) | $= \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q)$                                                                  |                                                                                                                                                              |

$$= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$$

at the layered network  $(F(A,X)=A+A \times X, B \text{ the paths } V^+, \alpha=[\text{wrap}, \text{cons}]): \frac{F(1,\exists)}{\exists} P(\alpha) \text{ est}(R)=[\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$   
 $P(\text{cons}) \text{ est}(R)$

$$= \frac{F(\exists,1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists,1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A,X) & = & A & + & A \times X \\ & & | & & | \times | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A',X') & = & A' & + & A' \times X' \end{array} \quad (8.2e)$$

u  $F(R,S)$  v holds iff one of:

- u, v are both inl and  $d \leq R \leq d'$ : the leaf moves by R;
- u, v are both inr and  $p_1 \leq R \leq q_1 \wedge p_2 \leq S \leq q_2$ : the vertex moves by R, the rest by S.
- An inl never relates to an inr:  $R+S \triangleq [R \text{ inl}, S \text{ inr}]$  returns to the summand it came from, and  $F(1)=1$  allows no crossing.
- $F(\exists,\exists)$  relates  $\text{inr}(\{a,b\},\{p,q\})$  to  $\text{inr}(a,p), \text{inr}(a,q), \text{inr}(b,p), \text{inr}(b,q)$ , and  $\text{inl}(\{a,b\})$  to  $\text{inl}(a), \text{inl}(b)$ .

## §8.3. Implementing thin

(8.3a)

|   | name           | type                           | definition                                                                                                                                                                                            | meaning                                                                                        |
|---|----------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 | setify         | $[A] \rightarrow PA$           | $(\text{setify } \exists)(xs, x) \iff x \in xs$                                                                                                                                                       | the set of a list's elements                                                                   |
| 2 | cup            | $PA \times PA \rightarrow PA$  | $\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$                                                                                                                          | the union of two sets                                                                          |
| 3 | cp(F)          | $F(PA) \rightarrow PFA$        | $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$                                                                                                                                                  | every F-shaped value whose parts are chosen from the given sets                                |
| 4 | sort( $\leq$ ) | $PA \rightarrow LA$            | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$                                                                                                                              | read a set back as one of its $\leq$ -ordered listings                                         |
| 5 | bump(Q)        | $A \times [A] \rightarrow [A]$ | $(a, []) \mapsto [a]$ , $\text{bump}(Q)((a, [b] \# xs), ys) \iff Q(a, b) \wedge ys = [a] \# xs \vee Q(b, a) \wedge ys = [b] \# xs \vee \neg Q(a, b) \wedge \neg Q(b, a) \wedge ys = [a] \# [b] \# xs$ | put $a$ in front of a thinned list, dropping whichever of $a$ and the old head the other beats |
| 6 | thinlist(Q)    | $[A] \rightarrow [A]$          | $\text{thinlist}(Q) = \lambda [nil, \text{bump}(Q)]$                                                                                                                                                  | thin a list in one pass, bumping each element into the thinned rest                            |
| 7 |                |                                | $\text{IsThinlist}(Q, \text{thinlist}) \iff \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q)$                                                | what a list implementation of $\text{thin}(Q)$ must do: only drop elements, and thin the set   |
| 8 | minlist(Q)     | $[A] \rightarrow A$            | $\text{minlist}(Q) \triangleq \text{setify est}(Q)$                                                                                                                                                   | a $Q$ -least element of the list                                                               |

(8.3b)

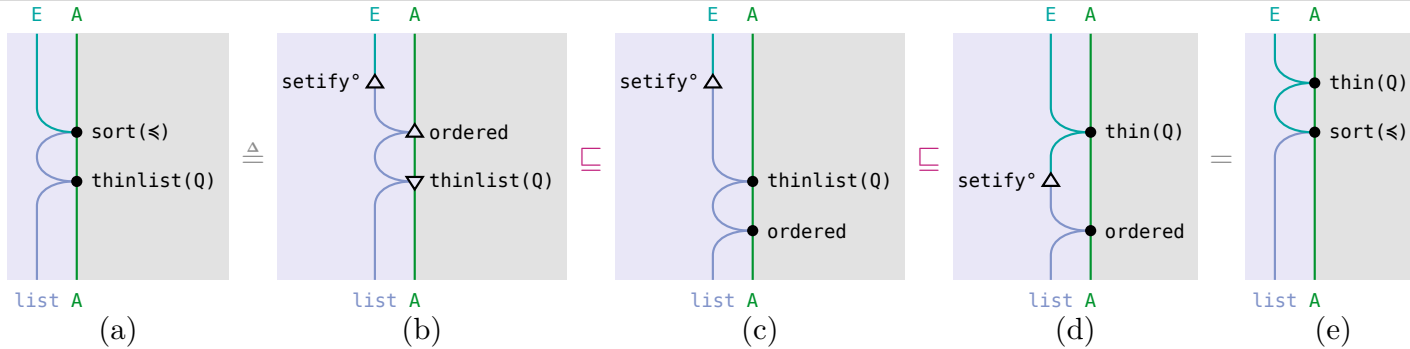
**Binary thinning data:**  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $\leq$  a connected preorder with both  $f_1, f_2$  monotonic on  $\leq$ ;  $R$  a preorder.

(8.3c)

|   | the law                                                                                                                                                                                                                                               | what it says                                                                                                       |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{preorder}(Q) \wedge \text{connected}(Q) \implies \text{thinlist}(Q)$<br>$([a] \# xs, ys) \iff (\exists w. \text{minlist}(Q)([a] \# xs, w) \wedge ys = [w])$ (8.5)                                                                              | what thinning should come to when it can: one element                                                              |
| 2 | $\text{preorder}(Q) \implies \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q)$<br>$\text{sort}(\leq)$ (8.6) — (8.6) $\text{preorder}(Q) \implies \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| 3 | $\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)                                                                                                                                                                                | a minimum of the sorted list is a minimum of the set                                                               |
| 4 | $\text{sort}(g \leq g^\circ) \text{ list}(g) \sqsubseteq P(g) \text{ sort}(\leq)$ (8.8)                                                                                                                                                               | shunt a function through a sort                                                                                    |
| 5 | $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9)                                                                                                                                                                       | filtering a sorted list sorts the restricted set                                                                   |
| 6 | $\text{preorder}(\leq) \wedge \text{connected}(\leq) \implies (\text{sort}(\leq) \times \text{sort}(\leq))$<br>$\text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)                                                                          | merging two sorted lists sorts their union                                                                         |
| 7 | $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$<br>(8.11), $FX = L + E \times X$                                                                                                                                 | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

**preorder(Q)  $\implies$  sort( $\leq$ ) thinlist(Q)  $\sqsubseteq$  thin(Q) sort( $\leq$ )**

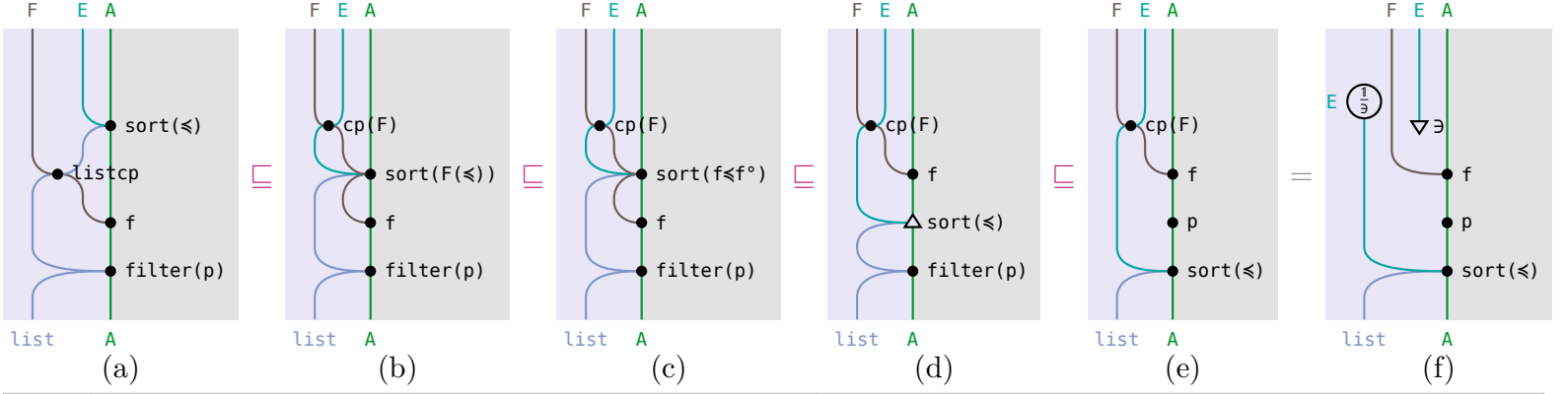
(8.6)

sorting the set then thinning the list by  $Q$  lists, sorted, a  $Q$ -thinning of the set —  $Q$  a preorder.

|     |                                                                       |                                                                                                    |
|-----|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| (a) | $\text{sort}(\leq) \text{ thinlist}(Q)$                               | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$                           |
| (b) | $\triangleq \text{setify}^\circ \text{ ordered thinlist}(Q)$          | $\text{ordered thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered}$                        |
| (c) | $\sqsubseteq \text{setify}^\circ \text{ thinlist}(Q) \text{ ordered}$ | $\text{preorder}(Q)$                                                                               |
| (d) | $\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}$     | $\implies \text{setify}^\circ \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ setify}^\circ$ |
| (e) | $= \text{thin}(Q) \text{ sort}(\leq)$                                 |                                                                                                    |

$$F(\leq) \quad f \sqsubseteq f \leq \implies F(\text{sort}(\leq)) \quad \text{listcp} \quad \text{list}(f) \quad \text{filter}(p) \sqsubseteq \frac{F(\exists) \quad f \quad p}{\exists} \quad \text{sort}(\leq)$$

Lemma 8.1

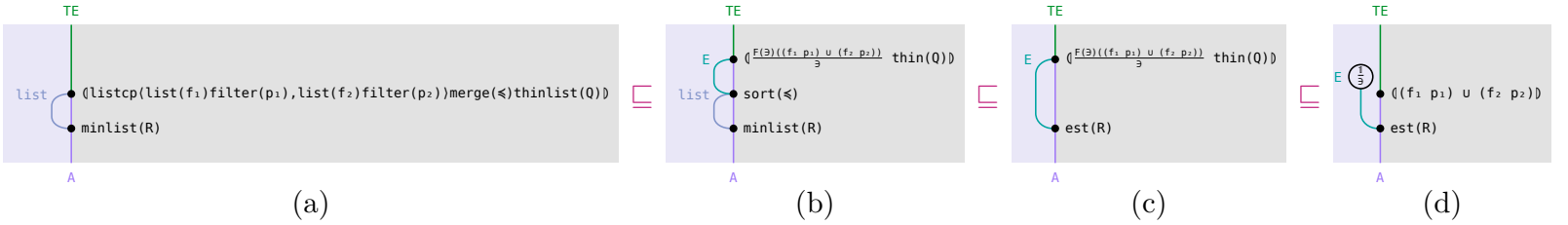
one sorted list built from sorted arguments, instead of a set built and then sorted —  $f : FA \rightarrow A$  monotonicon  $\leq$ .

|     |                                                                                                          |                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| (a) | $F(\text{sort}(\leq)) \quad \text{listcp} \quad \text{list}(f) \quad \text{filter}(p)$                   | $F(\text{sort}(\leq)) \quad \text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sort}(F(\leq))$ |
| (b) | $\sqsubseteq \text{cp}(F) \quad \text{sort}(F(\leq)) \quad \text{list}(f) \quad \text{filter}(p)$        | $F(\leq) \quad f \sqsubseteq f \leq$                                                           |
| (c) | $\sqsubseteq \text{cp}(F) \quad \text{sort}(f \leq f^\circ) \quad \text{list}(f) \quad \text{filter}(p)$ | $\text{sort}(g \leq g^\circ) \quad \text{list}(g) \sqsubseteq P(g) \quad \text{sort}(\leq)$    |
| (d) | $\sqsubseteq \text{cp}(F) \quad P(f) \quad \text{sort}(\leq) \quad \text{filter}(p)$                     | $\text{sort}(\leq) \quad \text{filter}(p) \sqsubseteq E(p) \quad \text{sort}(\leq)$            |
| (e) | $\sqsubseteq \text{cp}(F) \quad E(f \quad p) \quad \text{sort}(\leq)$                                    | $\frac{S}{\exists} \quad E(R) = \frac{S \quad R}{\exists}$                                     |
| (f) | $= \frac{F(\exists) \quad f \quad p}{\exists} \quad \text{sort}(\leq)$                                   |                                                                                                |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad f_1 \quad p_1 \sqsubseteq f_1 \quad p_1 \quad Q \wedge F(Q) \quad f_2 \quad p_2 \sqsubseteq f_2 \quad p_2 \quad Q \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \implies (\text{listcp} \quad (\text{list}(f_1) \quad \text{filter}(p_1), \text{list}(f_2) \quad \text{filter}(p_2)) \quad \text{merge}(\leq) \quad \text{thinlist}(Q)) \quad \text{minlist}(R) \sqsubseteq \frac{Q((f_1 \quad p_1) \cup (f_2 \quad p_2))}{\exists} \quad \text{est}(R)$$

Theorem 8.2

a fold on sorted lists of partial solutions, thinned at every step, refines the thinning specification — at binary thinning data.

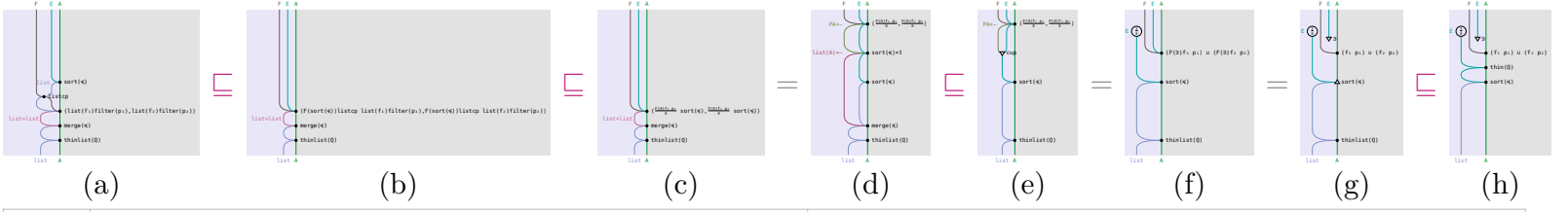


|     |                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(\text{listcp} \quad (\text{list}(f_1) \quad \text{filter}(p_1), \text{list}(f_2) \quad \text{filter}(p_2)) \quad \text{merge}(\leq) \quad \text{thinlist}(Q)) \quad \text{minlist}(R)$ | $(8.3d) \quad \text{preorder}(Q) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \implies F(\text{sort}(\leq)) \quad \text{listcp} \quad (\text{list}(f_1) \quad \text{filter}(p_1), \text{list}(f_2) \quad \text{filter}(p_2)) \quad \text{merge}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \frac{F(\exists) \quad ((f_1 \quad p_1) \cup (f_2 \quad p_2))}{\exists} \quad \text{thin}(Q) \quad \text{sort}(\leq)$ |
| (b) | $\sqsubseteq \frac{Q((f_1 \quad p_1) \cup (f_2 \quad p_2))}{\exists} \quad \text{thin}(Q) \quad \text{sort}(\leq) \quad \text{minlist}(R)$                                               | $\text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| (c) | $\sqsubseteq \frac{Q((f_1 \quad p_1) \cup (f_2 \quad p_2))}{\exists} \quad \text{thin}(Q) \quad \text{est}(R)$                                                                           | $F(Q) \quad S_1 \sqsubseteq S_1 \quad Q \wedge F(Q) \quad S_2 \sqsubseteq S_2 \quad Q \implies F(Q) \quad (S_1 \cup S_2) \sqsubseteq (S_1 \cup S_2) \quad Q$                                                                                                                                                                                                                                                                                                                                                               |
| (d) | $\sqsubseteq \frac{Q((f_1 \quad p_1) \cup (f_2 \quad p_2))}{\exists} \quad \text{est}(R)$                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

$$\text{preorder}(Q) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \quad f_1 \in f_1 \leq \wedge F(\leq) \quad f_2 \in f_2 \leq \implies F(\text{sort}(\leq)) \quad (8.3d)$$

$$\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$$

sorting the candidate set turns the thinning algebra into an algebra on sorted lists.



|     |                                                                                                                                                                                                            |                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q)$                                                  | $X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle$                                                                                  |
| (b) | $\sqsubseteq F(\text{sort}(\leq)) \text{ listcp} \text{ list}(f_1) \text{ filter}(p_1), F(\text{sort}(\leq)) \text{ listcp} \text{ list}(f_2) \text{ filter}(p_2) \text{ merge}(\leq) \text{ thinlist}(Q)$ | Lemma 8.1 $F(\leq) \quad f \sqsubseteq f \leq$                                                                                                 |
| (c) | $\sqsubseteq \langle \frac{F(\exists) f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists) f_2 p_2}{\exists} \text{ sort}(\leq) \rangle \text{ merge}(\leq) \text{ thinlist}(Q)$                         | $\implies F(\text{sort}(\leq)) \text{ listcp} \text{ list}(f) \text{ filter}(p) \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$ |
| (d) | $= \langle \frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q)$                            | $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$                                                                                 |
| (e) | $\sqsubseteq \langle \frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \rangle \text{ cup } \text{sort}(\leq) \text{ thinlist}(Q)$                                                    | $\text{preorder}(\leq) \wedge \text{connected}(\leq)$                                                                                          |
| (f) | $= \frac{(F(\exists) f_1 p_1) \cup (F(\exists) f_2 p_2)}{\exists} \text{ sort}(\leq) \text{ thinlist}(Q)$                                                                                                  | $\implies (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup } \text{sort}(\leq)$                          |
| (g) | $= \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ sort}(\leq) \text{ thinlist}(Q)$                                                                                                           | $\frac{R \cup S}{\exists} = \langle \frac{R}{\exists}, \frac{S}{\exists} \rangle \text{ cup}$                                                  |
| (h) | $\sqsubseteq \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$                                                                                                     | $R (S \cup T) = (R S) \cup (R T)$                                                                                                              |
|     |                                                                                                                                                                                                            | (8.6) $\text{preorder}(Q)$                                                                                                                     |
|     |                                                                                                                                                                                                            | $\implies \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$                                                 |

## §8.4. The knapsack problem

(8.4a)

|   | name                                                                     | type                                                                       | definition                                                                                                                                                                                                                                                                                                                                                                                                                                    | meaning                                                     |
|---|--------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | $\text{total}(f, xs)$                                                    | Int                                                                        | $\text{total}(f, xs) \triangleq \text{sum}(\text{list}(f, xs))$ , total $f = \text{list}(f) \text{ sum}$                                                                                                                                                                                                                                                                                                                                      | add up $f$ over the items of the list                       |
| 2 | $\text{subseq}$                                                          | $[A] \rightarrow [A]$                                                      | $xs \text{ subseq } ys \iff \text{subseq}(ys, xs)$                                                                                                                                                                                                                                                                                                                                                                                            | every subsequence of the list                               |
| 3 | $[\text{nil}, \text{cons}]$                                              | $F[\text{Item}] \rightarrow [\text{Item}]$                                 | $[\text{nil}, \text{cons}] = [\text{nil}, \text{cons}]$ ,<br>$[\text{nil}, \text{cons}](\text{l}(u)) = []$ ,<br>$[\text{nil}, \text{cons}](\text{r}((a, x))) = [a] \# x$                                                                                                                                                                                                                                                                      | keep the item: the list constructor                         |
| 4 | $[\text{nil}, \pi_2]$                                                    | $\text{Unit} + \text{Item} \times [\text{Item}] \rightarrow [\text{Item}]$ | $[\text{nil}, \pi_2](\text{l}(D)) = []$ , $[\text{nil}, \pi_2](\text{r}(p)) = \pi_2(p)$ , $[\text{nil}, \pi_2] = [\text{nil}, \pi_2]$                                                                                                                                                                                                                                                                                                         | drop the item: keep the tail                                |
| 5 | $\text{within}(w)$                                                       | $[\text{Item}] \rightarrow [\text{Item}]$                                  | $\text{within}(w)(x, y) \iff x = y \wedge \text{total}(\text{wt}, x) \leq w$                                                                                                                                                                                                                                                                                                                                                                  | the packings whose total weight is at most $w$              |
| 6 | $([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2]$ | $F[\text{Item}] \rightarrow [\text{Item}]$                                 | $([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2] \triangleq ([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2]$ ,<br>$([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2] = [\text{nil}, (\text{cons} \text{ within}(w)) \cup \pi_2]$ ,<br>$([\text{nil}, \text{cons}] \text{ within}(w))(u, r) \iff r = [\text{nil}, \text{cons}](u) \wedge \text{total}(\text{wt}, r) \leq w$ | at each item, keep it if the packing still fits, or drop it |
| 7 | $R$                                                                      | $[\text{Item}] \rightarrow [\text{Item}]$                                  | $R = \text{total vol} \geq \text{total vol}^\circ$ ,<br>$R(x, y) \iff \text{total}(\text{vol}, y) \leq \text{total}(\text{vol}, x)$                                                                                                                                                                                                                                                                                                           | $x$ is worth at least as much as $y$                        |
| 8 | $Q(\text{vol}, \text{wt})$                                               | $[\text{Item}] \rightarrow [\text{Item}]$                                  | $Q(\text{vol}, \text{wt}) = R \cap (\text{total wt} \leq \text{total wt}^\circ)$ ,<br>$Q(\text{vol}, \text{wt})(x, y) \iff \text{total}(\text{vol}, y) \leq \text{total}(\text{vol}, x) \wedge \text{total}(\text{wt}, x) \leq \text{total}(\text{wt}, y)$                                                                                                                                                                                    | $x$ is worth at least as much as $y$ and weighs no more     |

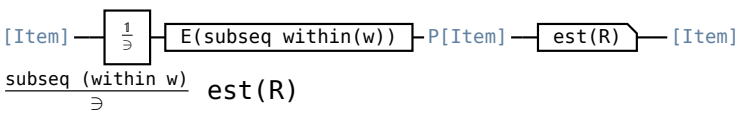
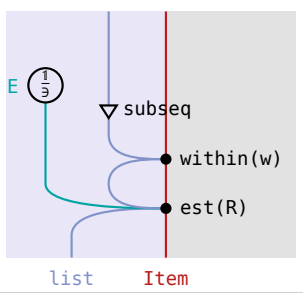
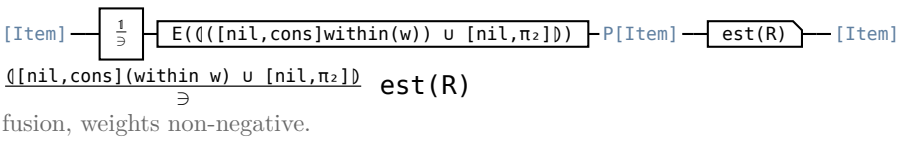
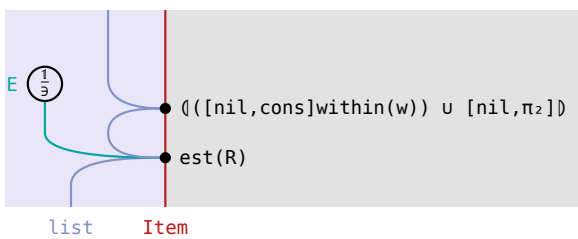
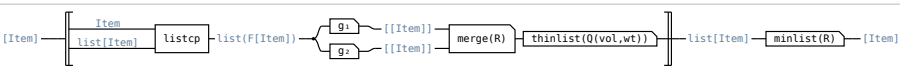
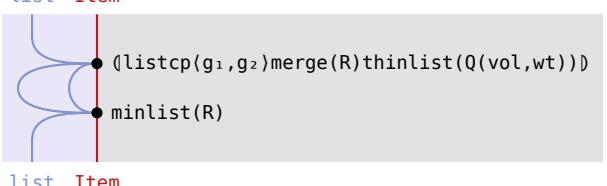
(8.4b)

| the law                                                                                                                                                                                                                                            | what it says                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w) R$<br>FALSE                                                                                                                                                         | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| 2 $F(Q(\text{vol}, \text{wt})) \text{ [nil, cons] within}(w) \sqsubseteq \text{[nil, cons] within}(w) Q(\text{vol}, \text{wt})$<br>$F(Q(\text{vol}, \text{wt})) \text{ [nil, } \pi_2] \sqsubseteq \text{[nil, } \pi_2] Q(\text{vol}, \text{wt})$ , | both halves are monotonic on Q once ties in value are broken by weight                         |
| 3 $F(R) \text{ [nil, cons] } \sqsubseteq \text{[nil, cons] } R$<br>$F(R) \text{ [nil, } \pi_2] \sqsubseteq \text{[nil, } \pi_2] R$                                                                                                                 | both algebras are monotonic on R itself, so R is the sort order P                              |

$$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \Rightarrow (\text{listcp } \langle g_1, g_2 \rangle \text{ merge}(R) \text{ thinlist}(Q(\text{vol}, \text{wt}))) \text{ minlist}(R) \sqsubseteq \text{est}(R) \quad (8.4c)$$

$\frac{\text{subseq within}(w)}{\supset}$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                        |    |
|                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|  <p><math>\sqsubseteq</math> binary thinning theorem, at <math>P \triangleq R</math>, F linear, Q from <math>F(Q(\text{vol}, \text{wt}))</math><br/> <math>\text{[nil, cons] within}(w) \sqsubseteq \text{[nil, cons] within}(w) Q(\text{vol}, \text{wt})</math>,<br/> <math>F(Q(\text{vol}, \text{wt})) \text{ [nil, } \pi_2] \sqsubseteq \text{[nil, } \pi_2] Q(\text{vol}, \text{wt})</math></p> |  |
| <p><math>\text{Q[nil, cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \text{ minlist}(R)</math><br/> <math>= \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] - [\text{nil, cons}] =</math><br/> <math>[\text{nil, cons}], [\text{nil, } \pi_2](l(D)) = [], [\text{nil, } \pi_2](r(p)) = \pi_2(p);</math><br/> <math>\text{minlist}(R)</math> is head, packings coming out in descending value</p>                                   |                                                                                       |

## §8.5. The paragraph problem

(8.5a)

|    | name                                             | type                                     | definition                                                                                                                                                                                                                                                    | meaning                                                                          |
|----|--------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1  | Line                                             | Type                                     | $\text{Line} \triangleq L(\text{Word})$                                                                                                                                                                                                                       | a line is a non-empty list of words                                              |
| 2  | Para                                             | Type                                     | $\text{Para} \triangleq L(\text{Line})$                                                                                                                                                                                                                       | a paragraph is a non-empty list of lines                                         |
| 3  | $\text{new}(a,p)$                                | Para                                     | $\text{new}(a,p) \triangleq [[a]]\#p$                                                                                                                                                                                                                         | start a new first line holding only the word a                                   |
| 4  | $\text{glue}(a)$                                 | $\text{Para} \rightarrow \text{Para}$    | $\text{glue}(a,[l]) = [[a]\#l], \text{glue}(a,[l]\#a') = [[a]\#l]\#a'$                                                                                                                                                                                        | put the word a at the front of the first line                                    |
| 5  | $[\text{wrap wrap},\text{new}]$                  | $F(\text{Para}) \rightarrow \text{Para}$ | $[\text{wrap wrap},\text{new}](l(a)) = [[a]], [\text{wrap wrap},\text{new}](r(q)) = \text{new}(\pi_1(q),\pi_2(q))$                                                                                                                                            | one word is a one-line paragraph; each further word starts a new line            |
| 6  | $[\text{wrap wrap},\text{glue}]$                 | $F(\text{Para}) \rightarrow \text{Para}$ | $[\text{wrap wrap},\text{glue}](l(a)) = [[a]], [\text{wrap wrap},\text{glue}](r(q)) = \text{glue}(\pi_1(q),\pi_2(q))$                                                                                                                                         | one word is a one-line paragraph; each further word joins the first line         |
| 7  | $[\text{wrap wrap},\text{new} \cup \text{glue}]$ | $F(\text{Para}) \rightarrow \text{Para}$ | $[\text{wrap wrap},\text{new} \cup \text{glue}] \triangleq [\text{wrap wrap},\text{new}] \cup [\text{wrap wrap},\text{glue}]$                                                                                                                                 | each further word either starts a new line or joins the first line               |
| 8  | partition                                        | $L(\text{Word}) \rightarrow \text{Para}$ | $\text{partition} \triangleq ([\text{wrap wrap},\text{new} \cup \text{glue}]) ,$<br>$\text{partition}([a],q) \iff q = [[a]] ,$<br>$\text{partition}([a]\#x,q) \iff (\exists p. \text{partition}(x,p) \wedge (q = \text{new}(a,p) \vee q = \text{glue}(a,p)))$ | every way of breaking the words into lines                                       |
| 9  | width                                            | $\text{Line} \rightarrow \text{Int}$     | $\text{width}[a] = \text{len}(a), \text{width}([a]\#x) = (\text{len}(a) + \text{width}(x)) + 1$                                                                                                                                                               | the lengths of the words plus one space between neighbours                       |
| 10 | head                                             | $\text{Para} \rightarrow \text{Line}$    | $\text{head}[l] = l, \text{head}([l]\#a) = l$                                                                                                                                                                                                                 | the first line of the paragraph                                                  |
| 11 | $\text{fits}(w)$                                 | $P(\text{Para})$                         | $\text{fits}(w,[l]) = \text{width}(l) \leq w, \text{fits}(w,[l]\#a) = \text{width}(l) \leq w \wedge \text{fits}(w,a)$                                                                                                                                         | every line is at most w wide                                                     |
| 12 | $\text{fits}(w)$                                 | $\text{Para} \rightarrow \text{Para}$    | $\text{fits}(w)(p,q) \iff p = q \wedge \text{fits}(w,p)$                                                                                                                                                                                                      | the paragraphs whose every line is at most w wide                                |
| 13 | $\text{ok}(w)$                                   | $\text{Para} \rightarrow \text{Para}$    | $\text{ok}(w)(p,q) \iff p = q \wedge \text{width}(\text{head}(p)) \leq w$                                                                                                                                                                                     | the paragraphs whose first line is at most w wide                                |
| 14 | $\text{sqr}(n)$                                  | Int                                      | $\text{sqr}(n) \triangleq nn$                                                                                                                                                                                                                                 | n squared                                                                        |
| 15 | $\text{waste}(w)$                                | $\text{Para} \rightarrow \text{Int}$     | $\text{waste}(w,[l]) = 0, \text{waste}(w,[l]\#a) = \text{sqr}(w - \text{width } l) + \text{waste}(w,a)$                                                                                                                                                       | the squared white space left on every line but the last, added up                |
| 16 | R                                                | $\text{Para} \rightarrow \text{Para}$    | $R = \text{waste } w \leq \text{waste } w^\circ, R(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q)$                                                                                                                                                        | p wastes no more than q                                                          |
| 17 | $Q(\text{len},w)$                                | $\text{Para} \rightarrow \text{Para}$    | $Q(\text{len},w) = R \cap (\text{head head}^\circ), Q(\text{len},w)(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q) \wedge \text{head}(p) = \text{head}(q)$                                                                                                | p wastes no more than q and has the same first line                              |
| 18 | $\text{start}(a)$                                | [Para]                                   | $\text{start}(a) \triangleq [[[a]]]$                                                                                                                                                                                                                          | the one candidate for a single word: the paragraph of one line holding that word |

(8.5b)

| the law                                                                                                                                                                                                                                                                                                                                                   | what it says                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                                                                                                                                                                               | <b>glue</b> is not monotonic on <b>R</b> : the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| 2 $F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)$<br>$(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)$<br>$\text{cons monotonic on collect} \leq \text{collect}^\circ.$ | both halves are monotonic on <b>Q</b> once ties in waste are broken by the first line                                                   |
| 3 $\text{merge}(\tau) = \text{cat}$                                                                                                                                                                                                                                                                                                                       | $\tau$ needs no sorting at all                                                                                                          |
| 4 $F(\tau) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} \tau$<br>$F(\tau) \text{ [wrap wrap, glue]} \sqsubseteq \text{[wrap wrap, glue]} \tau$                                                                                                                                                                                            | both algebras are monotonic on $\tau$ , so $\tau$ is the sort order <b>P</b>                                                            |

$$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \Rightarrow (Q(\text{listcp } \langle g_1, g_2 \rangle \text{ merge}(\tau) \text{ thinlist}(Q(\text{len}, w)))) \text{ minlist}(R) \sqsubseteq \frac{\text{partition fits}(w)}{\tau} \text{ est}(R) \quad (8.5c)$$

a paragraph laid out as a fold that thins the layouts kept at each word

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hinze–Marsden |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $\frac{\text{partition L(fits } w)}{\tau} \text{ est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\text{[wrap wrap, new] } \cup \text{ ([wrap wrap, glue] ok}(w))})} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $= \frac{Q(\text{[wrap wrap, new] } \cup \text{ ([wrap wrap, glue] ok}(w))})}{\tau} \text{ est}(R)$ <p>fusion, every word fits on a line by itself.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\text{[wrap wrap, new] } \cup \text{ ([wrap wrap, glue] ok}(w))})} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $= \frac{Q(\text{[wrap wrap, new] } \cup \text{ ([wrap wrap, glue] ok}(w))})}{\tau} \text{ est}(R)$ <p>the algebra as <math>(f_1 p_1) \cup (f_2 p_2)</math>, <math>p_1 \triangleq 1</math> — binary thinning theorem.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| $\text{L}(\text{Word}) \xrightarrow{\text{F}(\text{list}(\text{L}(\text{Line})))} \text{listcp} \text{ list}(\text{F}(\text{L}(\text{Line}))) \xrightarrow{\begin{smallmatrix} 0_1 \\ 0_2 \end{smallmatrix}} \begin{smallmatrix} \text{L}(\text{Line}) \\ \text{L}(\text{Line}) \end{smallmatrix} \xrightarrow{\text{merge}(\tau)} \text{thinlist}(Q(\text{len}, w)) \xrightarrow{\text{list}(\text{L}(\text{Line}))} \text{minlist}(R) \xrightarrow{\text{L}(\text{Line})}$ $Q(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R)$ <p>binary thinning theorem, at <math>P \triangleq T</math> with <math>\text{merge } \tau = \text{cat}</math> (<math>\text{merge}(\tau) = \text{cat}</math>), <math>Q</math> from <math>F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)</math>, <math>(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)</math></p> |               |
| $Q(\text{[start, cpr } \langle h_1, h_2 \rangle \text{ cat thinlist}(Q)]) \text{ minlist}(R)$ <p><math>\text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i</math> along the coproduct — <math>\text{[wrap wrap, new]}</math></p> $= (\text{l}(a)) = [[a]], \text{[wrap wrap, new]}(r(q)) = \text{new}(\pi_1(q), \pi_2(q)), \text{[wrap wrap, glue]}(\text{l}(a)) = [[a]], \text{[wrap wrap, glue]}(r(q)) = \text{glue}(\pi_1(q), \pi_2(q))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |

## §8.6. Bitonic tours

(8.6a)

|    | name                                                                   | type                                                                  | definition                                                                                                                                                                                                                                                                                                                                                             | meaning                                                                             |
|----|------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1  | Journey(City)                                                          | Type                                                                  | $\text{Journey}(\text{City}) \triangleq \mu F$                                                                                                                                                                                                                                                                                                                         | a list of at least two cities, the last two held as a pair                          |
| 2  | Tour(City)                                                             | Type                                                                  | $\text{Tour}(\text{City}) \triangleq \text{Journey}(\text{City}) \times \text{Journey}(\text{City})$                                                                                                                                                                                                                                                                   | the outward journey and the return journey                                          |
| 3  | head                                                                   | $\text{Journey}(\text{City}) \rightarrow \text{City}$                 | $\text{head}(\text{wrap}(p)) = \pi_1(p), \text{head}([a]\#a') = a$                                                                                                                                                                                                                                                                                                     | the first city of the journey                                                       |
| 4  | next                                                                   | $\text{Journey}(\text{City}) \rightarrow \text{City}$                 | $\text{next}(\text{wrap}(p)) = \pi_2(p), \text{next}([a]\#a') = \text{head}(a')$                                                                                                                                                                                                                                                                                       | the second city of the journey                                                      |
| 5  | $[a]\#\text{tail}(\cdot)$                                              | $\text{Journey}(\text{City}) \rightarrow \text{Journey}(\text{City})$ | $[a]\#\text{tail}(\text{wrap}(p)) = \text{wrap}(a, \pi_2(p)), [a]\#\text{tail}([a']\#a'') = [a]\#a''$                                                                                                                                                                                                                                                                  | replace the first city of the journey by a                                          |
| 6  | outcost                                                                | $\text{Journey}(\text{City}) \rightarrow \text{Int}$                  | $\text{outcost}(\text{wrap}(p)) = \text{tc}(p), \text{outcost}([a]\#a') = \text{tc}(a, \text{head}(a')) + \text{outcost}(a')$                                                                                                                                                                                                                                          | the cost of travelling the journey forwards                                         |
| 7  | incost                                                                 | $\text{Journey}(\text{City}) \rightarrow \text{Int}$                  | $\text{incost}(\text{wrap}(p)) = \text{tc}(\pi_2(p), \pi_1(p)), \text{incost}([a]\#a') = \text{tc}(\text{head}(a'), a) + \text{incost}(a')$                                                                                                                                                                                                                            | the cost of travelling the journey backwards                                        |
| 8  | cost(t)                                                                | Int                                                                   | $\text{cost}(t) \triangleq \text{outcost}(\pi_1(t)) + \text{incost}(\pi_2(t))$                                                                                                                                                                                                                                                                                         | the outward journey forwards plus the return journey backwards                      |
| 9  | start                                                                  | $\text{City} \times \text{City} \rightarrow \text{Tour}(\text{City})$ | $\text{start}(a) \triangleq (\text{wrap}(a), \text{wrap}(a))$                                                                                                                                                                                                                                                                                                          | the tour of the last two cities, both journeys being that one edge                  |
| 10 | dropl(a,t)                                                             | Tour(City)                                                            | $\text{dropl}(a, t) \triangleq ([a]\#\text{tail}(\pi_1(t)), [a]\#\pi_2(t))$                                                                                                                                                                                                                                                                                            | a replaces the first city of the outward journey and is put in front of the return  |
| 11 | dropr(a,t)                                                             | Tour(City)                                                            | $\text{dropr}(a, t) \triangleq ([a]\#\pi_1(t), [a]\#\text{tail}(\pi_2(t)))$                                                                                                                                                                                                                                                                                            | a is put in front of the outward journey and replaces the first city of the return  |
| 12 | [start, dropl]                                                         | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | $[\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q))$                                                                                                                                                                                                                                          | start at the last two cities, then add each city by dropl                           |
| 13 | [start, dropr]                                                         | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | $[\text{start}, \text{dropr}](l(p)) = \text{start}(p), [\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q))$                                                                                                                                                                                                                                          | start at the last two cities, then add each city by dropr                           |
| 14 | [start, dropl u dropr]                                                 | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | $[\text{start}, \text{dropl} \cup \text{dropr}] \triangleq [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}], [\text{start}, \text{dropl} \cup \text{dropr}](u, t) = t = [\text{start}, \text{dropl}](u) \vee t = [\text{start}, \text{dropr}](u)$                                                                                                        | each further city is added by dropl or by dropr                                     |
| 15 | $\llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket$ | $\mu F \rightarrow \text{Tour}(\text{City})$                          | $\llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket \triangleq \llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket$                                                                                                                                                                                                                 | every bitonic tour of the cities                                                    |
| 16 | R                                                                      | $\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$       | $R = \text{cost} \leq \text{cost}^\circ, R(t, t') \iff \text{cost}(t) \leq \text{cost}(t')$                                                                                                                                                                                                                                                                            | t costs no more than t'                                                             |
| 17 | next2(t)                                                               | $\text{City} \times \text{City}$                                      | $\text{next2}(t) \triangleq (\text{next}(\pi_1(t)), \text{next}(\pi_2(t)))$                                                                                                                                                                                                                                                                                            | the second cities of both journeys                                                  |
| 18 | head2(t)                                                               | $\text{City} \times \text{City}$                                      | $\text{head2}(t) \triangleq (\text{head}(\pi_1(t)), \text{head}(\pi_2(t)))$                                                                                                                                                                                                                                                                                            | the first cities of both journeys                                                   |
| 19 | Q                                                                      | $\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$       | $Q = R \cap (\text{next2} \text{ next2}^\circ) \cap (\text{head2} \text{ head2}^\circ), Q(t, t') \iff \text{cost}(t) \leq \text{cost}(t') \wedge (\text{next}(\pi_1(t)) = \text{next}(\pi_1(t')) \wedge \text{next}(\pi_2(t)) = \text{next}(\pi_2(t'))) \wedge (\text{head}(\pi_1(t)) = \text{head}(\pi_1(t')) \wedge \text{head}(\pi_2(t)) = \text{head}(\pi_2(t')))$ | t costs no more than t', and both journeys of t and t' share their first two cities |

(8.6b)

| the law                                                                                                                                          | what it says                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ drop} \sqsubseteq \text{drop} \ R \text{ FALSE}$<br>$(1 \times R) \text{ drop} \sqsubseteq \text{drop} \ R \text{ FALSE}$ | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| 2 $F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$<br>$F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$ ,       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |
| 3 $F(T) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ T$<br>$F(T) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ T$         | both algebras are monotonic on T, so T is the sort order P                                                         |

$$(\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(\text{[start, drop]} \sqcup \text{drop})}{\exists} \text{ est}(R) \quad (8.6c)$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\mu F \text{ -- } \left[ \frac{1}{\exists} \right] \text{ -- } P(\mu F) \text{ -- } E(\text{[start, drop]} \sqcup \text{drop}) \text{ -- } P(\text{Journey}(\text{City})^2) \text{ -- } \text{est}(R) \text{ -- } \text{Journey}(\text{City})$ $\text{tour} \sqsupseteq \text{est}(R)$                                                                                                                                                                                                                                                   |               |
| $\mu F \text{ -- } \left[ \frac{1}{\exists} \right] \text{ -- } P(\mu F) \text{ -- } E(\text{[start, drop]} \sqcup \text{[start, drop]}) \text{ -- } P(\text{Journey}(\text{City})^2) \text{ -- } \text{est}(R) \text{ -- } \text{Journey}(\text{City})$ $= \frac{(\text{[start, drop]} \sqcup \text{drop})}{\exists} \text{ est}(R)$ $(\text{[start, drop]} \sqcup \text{drop}) \triangleq (\text{[start, drop]} \sqcup \text{drop})$                                                                                                    |               |
| $\mu F \text{ -- } \left[ \frac{1}{\exists} \right] \text{ -- } P(\mu F) \text{ -- } E(\text{[start, drop]} \sqcup \text{[start, drop]}) \text{ -- } P(\text{Journey}(\text{City})^2) \text{ -- } \text{est}(R) \text{ -- } \text{Journey}(\text{City})$ $= \frac{(\text{[start, drop]} \sqcup \text{drop})}{\exists} \text{ est}(R)$ $(\text{[start, drop]} \sqcup \text{drop}) \triangleq (\text{[start, drop]} \sqcup \text{drop})$                                                                                                    |               |
| $\mu F \text{ -- } \left[ \frac{1}{\exists} \right] \text{ -- } P(\mu F) \text{ -- } E(\text{[start, drop]} \sqcup \text{[start, drop]}) \text{ -- } P(\text{Journey}(\text{City})^2) \text{ -- } \text{est}(R) \text{ -- } \text{Journey}(\text{City})$ $= \frac{(\text{[start, drop]} \sqcup \text{drop})}{\exists} \text{ est}(R)$ $(\text{[start, drop]} \sqcup \text{drop}) \triangleq (\text{[start, drop]} \sqcup \text{drop})$                                                                                                    |               |
| $(\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)) \text{ minlist}(R)$ <p>binary thinning theorem, at <math>P \triangle T</math> with merge <math>T = \text{cat}</math>, <math>Q</math> from <math>F(Q)</math></p> $\text{[start, drop]} \sqsubseteq \text{[start, drop]} \ Q, F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$                                                                                                                                                                                         |               |
| $(\text{[start wrap, cpr } (\text{list}(\text{drop}_1), \text{list}(\text{drop}_2)) \text{ cat thinlist}(Q)) \text{ minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \text{ -- }$ $= [\text{start, drop}_1](l(p)) = \text{start}(p), [\text{start, drop}_1](r(q)) = \text{drop}_1(\pi_1(q), \pi_2(q)), [\text{start, drop}_2](l(p)) = \text{start}(p), [\text{start, drop}_2](r(q)) = \text{drop}_2(\pi_1(q), \pi_2(q)); \text{quadratic, two tours added per step}$ |               |